

CHE 555 – Homework 1

Full Solution

REGRESSION ANALYSIS REPORT

CHE 555 – Homework 1

PROBLEM 1

SETUP

Curve: $y = 2x^2 + 3x + 1$

The distance from a point $(x, y(x))$ to the origin is

$$D(x) = \sqrt{x^2 + y(x)^2}$$

Minimising D^2 :

$$D^2(x) = x^2 + (2x^2 + 3x + 1)^2$$

$$dD^2(x^*)/dx = 2x + 2(2x^2 + 3x + 1)(4x + 3) = 0$$

$$2x + 2(2x^2 + 3x + 1)(4x + 3) = 0 \rightarrow 8x^3 + 22x^2 + 19x + 3 = 0$$

The roots of this cubic equations were solved numerically using the Newton-Raphson method.

Optimal x	= -0.36380427
$y(x^*)$	= $2(-0.363804)^2 + 3(-0.363804) + 1$
	= 0.17329428
$D^2(x^*)$	= $(-0.363804)^2 + (0.173294)^2$
	= 0.16238446
$D(x^*)$	= $\sqrt{0.16238446} = 0.40296955$

ANSWER

Nearest point : (-0.363804, 0.173294)

Distance : 0.402970

PROBLEM 2

Variables : x = side length of square base (cm)
 h = height (cm)

$$\text{Volume constraint : } x^2 \cdot h = 1000 \rightarrow h = 1000 / x^2$$

Surface area (no top lid) :

$$\begin{aligned} S(x) &= x^2 + 4 \cdot x \cdot h \\ &= x^2 + 4 \cdot x \cdot (1000/x^2) \\ &= x^2 + 4000/x \end{aligned}$$

ANALYTICAL SOLUTION

$$\begin{aligned} dS/dx &= 2x - 4000/x^2 = 0 \\ 2x^3 &= 4000 \end{aligned}$$

$$x^3 = 2000$$

$$x^* = 2000^{(1/3)}$$

$$d^2S/dx^2 = 2 + 8000/x^3 > 0 \quad \text{for all } x > 0$$

NUMERICAL VALUES

$$x^* = 2000^{(1/3)} = 12.599210 \text{ cm}$$

$$h^* = 1000 / x^{*2} = 1000 / 12.599210^2 = 6.299605 \text{ cm}$$

$$S^* = (12.5992)^2 + 4000/12.5992 = 476.220316 \text{ cm}^2$$

$$d^2S/dx^2|_{x^*} = 6.0000 > 0$$

ANSWER

Optimal base side length : 12.5992 cm
 Optimal height : 6.2996 cm
 Minimum surface area : 476.2203 cm²

PROBLEM 3

Minimise: $f(x_1, x_2) = 4x_1 - x_1^2 - 12$

Subject to:

$$g_1 : 25 - x_1^2 - x_2^2 = 0$$

$$g_2 : 10x_1 - x_1^2 + 10x_2 - x_2^2 - 34 \geq 0$$

$$g_3 : (x_1-3)^2 + (x_2-1)^2 \geq 0$$

$$x_1, x_2 \geq 0$$

VARIABLE COUNT

Total Variables : 2 (x_1, x_2)

DOF ANALYSIS

Each independent equality constraint reduces DOF by 1.
 Equality constraints : 1
 DOF = total variables - equality constraints = 2 - 1 = 1

Number of independent variables : 1
 One valid choice : { x_1 }

EXPLANATION

E1 defines x_2 as a function of x_1 : $x_2 = \sqrt{(25 - x_1^2)}$ (with $x_2 \geq 0$).
 Once x_1 is chosen (satisfying $0 \leq x_1 \leq 5$), x_2 is fully determined.
 The inequality constraints further restrict the feasible range of x_1 but do not reduce the number of independent variables.
 $g_3 = (x_1-3)^2 + (x_2-1)^2 \geq 0$ is always satisfied (sum of squares).

ANSWER

Total variables : 2
 Independent variables : 1
 One possible choice : { x_1 }

PROBLEM 4

GIVEN

Cost per barrel : $C(P) = 50 + 0.1P + 9000/P$ (\$/barrel)
 Selling price : \$300 / barrel
 P = production rate (barrels/day)

PART A – Minimise cost per barrel

$$\begin{aligned}dC/dP &= 0.1 - 9000/P^2 = 0 \\ P^2 &= 9000 / 0.1 = 90000 \\ P^* &= \sqrt{90000} = 300 \text{ barrels/day}\end{aligned}$$

$$d^2C/dP^2 = 18000/P^3 > 0 \text{ for } P > 0$$

Numerical check:

$$\begin{aligned}P^* &= 300.0 \text{ barrels/day} \\ C(P^*) &= 50 + 0.1(300.0) + 9000/300.0 = 110.00 \text{ \$/barrel} \\ d^2C/dP^2|_{P^*} &= 0.000667 > 0\end{aligned}$$

PART B – Maximise daily profit

$$\begin{aligned}\text{Revenue per day} &: R(P) = 300 \cdot P \\ \text{Total cost / day} &: TC(P) = C(P) \cdot P = 50P + 0.1P^2 + 9000 \\ \text{Profit / day} &: \Pi(P) = R - TC = 300P - 50P - 0.1P^2 - 9000 \\ &= 250P - 0.1P^2 - 9000\end{aligned}$$

$$\begin{aligned}d\Pi/dP &= 250 - 0.2P = 0 \\ P^* &= 250 / 0.2 = 1250 \text{ barrels/day}\end{aligned}$$

$$d^2\Pi/dP^2 = -0.2 < 0$$

Numerical check:

$$\begin{aligned}P^* &= 1250 \text{ barrels/day} \\ \text{Revenue} &= 300 \times 1250 = \$375,000.00/\text{day} \\ \text{Total cost} &= 50(1250) + 0.1(1250)^2 + 9000 = \$227,750.00/\text{day} \\ \text{Profit } \Pi(P^*) &= \$375,000.00 - \$227,750.00 = \$147,250.00/\text{day}\end{aligned}$$

ANSWERS

- (A) Production minimising cost/barrel : $P = 300$ barrel/day $\rightarrow C = \$110.00/\text{barrel}$
(B) Production maximising daily profit: $P = 1250$ barrel/day $\rightarrow \Pi = \$147,250.00/\text{day}$

PROBLEM 5

DATA

x	y
10	1.00
20	1.26
30	1.86
40	3.31
50	7.08

MODELS TO FIT

$$\begin{aligned}\text{Model 1 : } y &= \exp(a + b \cdot x) && [\text{exp-linear}] \\ \text{Model 2 : } y &= \exp(a + b \cdot x + c \cdot x^2) && [\text{exp-quadratic}] \\ \text{Model 3 : } y &= a \cdot x^b && [\text{power law}]\end{aligned}$$

METHOD

Models 1 & 2 are linearised by taking $\ln y$ and fitting via OLS.
Model 3 is linearised as $\ln y = \ln a + b \cdot \ln x$ and fitted via OLS.
Full regression statistics (R^2 , adj- R^2 , SE, RMSE, F-test, t-tests, ANOVA table) are reported for each model in the sections below.

Regression details:

Model: $y = \exp(a + b \cdot x)$

Observations : 5
Parameters (p) : 2
DOF (error) : 3

— Goodness of Fit —

R^2 : 0.955527
 R^2 (adjusted) : 0.940703
SE of regression : 0.192228
RMSE : 0.148900
SSE : 0.110855
SSR : 2.381815
SST : 2.492670

— Overall F-Test —

F-statistic : 64.4574
F p-value : 0.00403481

— Parameter t-Tests (H_0 : param = 0, $\alpha = 0.05$) —

Param	Estimate	Std Err	t-stat	p-value	95% CI lower	95% CI upper
a	-0.662933	0.201611	-3.2882	0.046141	-1.304549	-0.021317 *
b	0.048804	0.006079	8.0285	0.00403481	0.029458	0.068149 *

(* p < 0.05)

— ANOVA Table —

Source	df	SS	MS	F	p-value
Regression	1	2.381815	2.381815	64.4574	0.00403481
Residual	3	0.110855	0.036952		
Total	4	2.492670			

Model: $y = \exp(a + b \cdot x + c \cdot x^2)$

Observations : 5
Parameters (p) : 3
DOF (error) : 2

— Goodness of Fit —

R^2 : 0.999968
 R^2 (adjusted) : 0.999936
SE of regression : 0.006314
RMSE : 0.003993
SSE : 0.000080
SSR : 2.492591
SST : 2.492670

— Overall F-Test —

F-statistic : 31265.9950
F p-value : 3.19826e-05

— Parameter t-Tests (H_0 : param = 0, $\alpha = 0.05$) —

Param	Estimate	Std Err	t-stat	p-value	95% CI lower	95% CI upper
a	-0.040266	0.013541	-2.9736	0.096929	-0.098528	0.017997
b	-0.004568	0.001032	-4.4264	0.0474371	-0.009008	-0.000128 *
c	0.000890	0.000017	52.7166	0.000359642	0.000817	0.000962 *

(* p < 0.05)

— ANOVA Table —

Source	df	SS	MS	F	p-value
Regression	2	2.492591	1.246295	31265.9950	3.19826e-05
Residual	2	0.000080	0.000040		
Total	4	2.492670			

Model: $y = a \cdot x^b$ [log-log space]

Observations : 5
Parameters (p) : 2
DOF (error) : 3

— Goodness of Fit —
 R^2 : 0.818733
 R^2 (adjusted) : 0.758311
SE of regression : 0.388089
RMSE : 0.300612
SSE : 0.451838
SSR : 2.040832
SST : 2.492670

— Overall F-Test —
F-statistic : 13.5502
F p-value : 0.0347317

— Parameter t-Tests (H_0 : param = 0, $\alpha = 0.05$) —

Param	Estimate	Std Err	t-stat	p-value	95% CI lower	95% CI upper
ln a	-2.863028	1.010440	-2.8334	0.0660019	-6.078700	0.352644
b	1.123962	0.305337	3.6811	0.0347317	0.152245	2.095680 *

(* p < 0.05)

— ANOVA Table —

Source	df	SS	MS	F	p-value
Regression	1	2.040832	2.040832	13.5502	0.0347317
Residual	3	0.451838	0.150613		
Total	4	2.492670			

Original-scale R^2 for Model: $y = a \cdot x^b$: 0.726703 (adj R^2 : 0.635604)

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MODEL COMPARISON SUMMARY

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Model	R^2	adj R^2	RMSE	F-stat	F p-val
exp_linear	0.955527	0.940703	0.148900	64.4574	0.00403481
exp_quadratic	0.999968	0.999936	0.003993	31265.9950	3.19826e-05
power_law	0.726703	0.635604	0.300612	13.5502	0.0347317

Best model (highest adj R^2): exp_quadratic

FITTED VALUES vs. DATA

x	y_obs	exp-lin	exp-quad	power
10	1.0000	0.8395	1.0030	0.7596
20	1.2600	1.3677	1.2513	1.6554
30	1.8600	2.2282	1.8650	2.6111
40	3.3100	3.6300	3.3210	3.6079
50	7.0800	5.9136	7.0651	4.6364

RESIDUALS

x	exp-lin	exp-quad	power
10	0.1605	-0.0030	0.2404
20	-0.1077	0.0087	-0.3954
30	-0.3682	-0.0050	-0.7511
40	-0.3200	-0.0110	-0.2979
50	1.1664	0.0149	2.4436

CONCLUSION

The exp-quadratic model has the highest adjusted R^2
among the three candidates.
Best model: exp-quadratic

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END OF REPORT

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Plot

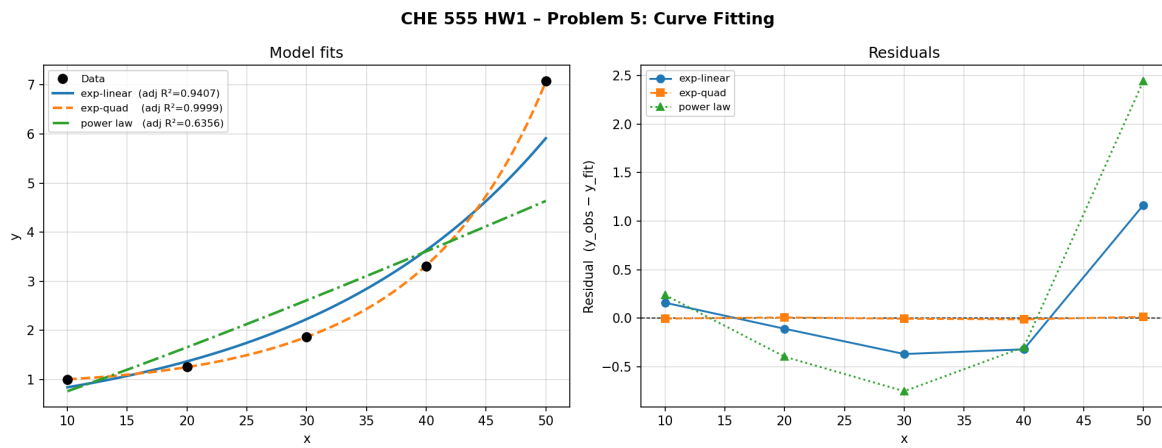


Figure 1: Curve fitting comparison for Problem 5.